

```
[.....]
```

This tells us that the MBR table for the second RAID array is completely blank. Well, we've proved our hypothesis that the system fault was because of a blank MBR. It's time to run GRUB, to generate an MBR for the new location:

```
# /usr/sbin/grub-install /dev/cciss/c1d0
```

Let's make sure that the MBR is generated and written correctly for the first disk:

```
# hexdump -v /dev/cciss/c1d0 | less
```

```
00000000 48eb 0090 0000 0000 0000 0000 0000 0000
00000010 0000 0000 0000 0000 0000 0000 0000 0000
00000020 0000 0000 0000 0000 0000 0000 0000 0000
00000030 0000 0000 0000 0000 0000 0000 0000 0203
00000040 00ff 2000 0001 0000 0200 90fa f690 80c2
00000050 0275 80b2 59ea 007c 3100 8ec0 8ead bcd0
00000060 2000 a0fb 7c40 ff3c 0274 c288 be52 7d7f
00000070 34e8 f601 80c2 5474 41b4 aabb cd55 5a13
00000080 7252 8149 55fb 75aa a043 7c41 c084 0575
00000090 e183 7401 6637 4c8b be10 7c05 44c6 01ff
000000a0 8b66 441e c77c 1004 c700 0244 0001 8966
000000b0 085c 44c7 0006 6670 c031 4489 6604 4489
000000c0 b40c cd42 7213 bb05 7000 7dab 08b4 13cd
000000d0 0a73 c2f6 0f80 aa84 e900 008d 05be c67c
000000e0 ff44 6600 c031 f088 6640 4489 3104 88d2
000000f0 c1ca 02a2 e888 f488 8940 0844 c031 d088
00001000 e8c0 6602 0489 a166 7c44 3166 66d2 34f7
00001010 5488 660a d231 f766 0474 5488 890b 0c44
00001020 443b 7d08 8a3c 0d54 a2c0 8a06 0a4c c1fe
00001030 d108 6c8a 5a0c 748a bb0b 7000 c38e db31
00001040 01b8 cd02 7213 8c2a 8ec3 4806 607c b91e
00001050 0100 db8e f631 ff31 f3fc 1fa5 ff61 4226
00001060 be7c 7d85 40e8 eb00 be0e 7d8a 38e8 eb00
00001070 7d94 30e8 be00 7d99 2ae8 eb00 47fe
00001080 5552 2042 4700 6665 006d 6148 6472 4420
```

```
0000190 7369 006b 6552 6461 2000 7245
6f72 0072
00001a0 01bb b400 cd0e ac10 003c f475
00c3 0000
00001b0 0000 0000 0000 0000 0000 0000
0000 0280
00001c0 0001 8183 8020 0040 0000 0000
0010 8200
00001d0 8001 fe8e ffe0 0040 0010 ac80
087a 0000
00001e0 0000 0000 0000 0000 0000 0000
0000 0000
00001f0 0000 0000 0000 0000 0000 0000
0000 aa55
```

Yes, the GRUB loader is generated on the first RAID disk. Now we flush any disk caches and do a warm reboot of the system, meanwhile disconnecting our mini-boot ISO from the virtual CD slot:

```
# sync && reboot
```

After reboot, we can enjoy the restored and, more importantly, the completely working system.

Here, I've explained in detail how to recover a Linux system with the help of the integrated iLO software from Hewlett-Packard. It no longer matters if you're far away from office, so long as you're able to log in to the company LAN via VPN, and able to access the server's iLO home page. From here, you can quite easily perform non-trivial tasks like a full system reconfiguration of a working system, complete installation, or restoring after a crash. 

Resources

- [1] http://en.wikipedia.org/wiki/HP_Integrated_Lights-Out
- [2] <http://www.hp.com/servers/lights-out>
- [3] http://en.wikipedia.org/wiki/Lights_out_management
- [4] http://en.wikipedia.org/wiki/Out-of-band_management
- [5] <http://syslinux.zytor.com/wiki/index.php/Download>
- [6] http://en.wikipedia.org/wiki/Saving_Virtual_Ryan

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The author has specialised in Linux and FOSS technologies for more than a decade. His professional spheres of interest include, but are not limited to, robotics, embedded systems, statistics and algorithmic methods.

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About Author: Vipin Gupta is BE, RHCE, CEH, ASM, CCNA, MCSE, MCSA. He has got more than 17 years of experience in IT related fields. During his long career, he has been associated with various engineering colleges, polytechnics, private institutes & corporate sector.

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